

SQL Code

```
1  create database loan_approved;
2  use loan_approved;
3
4  -- Basic Sql Quries
5
6  -- 1. View the first 10 records from the table.
7  select * from loan_approved_clean limit 10;
8
9  -- 2. Count the total number of loan applications.
10 select count(*) as total_applications from loan_approved_clean;
11
12
13 -- 3. List all unique property areas.
14 select distinct(Property_Area) from loan_approved_clean ;
15
16 -- 4. Show all applicants who are self-employed and have an income above 5000.
17 select * from loan_approved_clean where Self_Employed = 'Yes' and ApplicantIncome >
5000;
18 select count(*) from loan_approved_clean where Self_Employed = 'Yes' and
ApplicantIncome > 5000;
19
20
21 -- 5. Find the total number of approved loans.
22 select count(*) from loan_approved_clean where Loan_Status = 'Y';
23
24 -- 6. Find average loan amount by education level.
25 select round(avg(LoanAmount),2)as avg_Loan_Amount from loan_approved_clean group by
Education;
26
27 -- 7. Find average total income (Applicant + Coapplicant) by marital status.
28 select round(avg(ApplicantIncome + CoapplicantIncome),2)as Total_Income from
loan_approved_clean group by Married;
29 select Married, round(avg(applicantincome + coapplicantincome),2) as avg_totalincome
from loan_approved_clean group by Married;
30
31 -- 8. Show average loan amount by credit history.
32 select avg(LoanAmount) from loan_approved_clean group by Credit_History;
33 select credit_history,round(avg(loanamount),2) as avg_loanamount from
loan_approved_clean group by credit_history;
34
35 -- 9. Find total applications and approval rate by gender.
36 select Gender,
37    count(*) as total_applications,
38    sum(case when Loan_Status = 'Y' then 1 else 0 end) as Approvals,
39    round(sum(case when Loan_Status = 'Y' then 1 else 0 end) /count(*)*100,2) as
Approval_Rate
40 from loan_approved_clean group by Gender;
41
42 -- 10. Show approval rate by property area.
43 select Property_Area,
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44 round(sum(case when Loan_Status = 'Y' then 1 else 0 end) * 100 / count(*),2) as
Approvel_Rate
45 from loan_approved_clean group by Property_Area;
46
47 -- 11. Show applicants who are graduates, not self-employed, and have loan amount
greater than 150.
48 select * from loan_approved_clean where Education = 'Graduate' and Self_Employed =
'No' and LoanAmount >150;
49
50 -- 12. Display approved loans from urban area with good credit history.
51 select * from loan_approved_clean where Property_Area = 'Urban' and Credit_History = 1
and Loan_Status = 'Y';
52
53 -- 13. List top 5 applicants with highest total income.
54 select ApplicantIncome ,(ApplicantIncome+ CoapplicantIncome) as Total_Income from
loan_approved_clean order by Total_Income desc limit 5;
55 select * , applicantincome+coapplicantincome as total_income from loan_approved_clean
order by total_income desc limit 5;
56
57 -- 14. Create derived columns for total income for each applicant.
58 select * ,(ApplicantIncome+ CoapplicantIncome) as Total_Income from
loan_approved_clean;
59
60 -- 15. Classify applicants into income groups (Low, Medium, High) based on applicant
income.
61 select Loan_Id, ApplicantIncome,
62 case
63     when ApplicantIncome > 3000 then 'Low'
64     when ApplicantIncome between 3000 and 7000 then 'Medium'
65     else 'High'
66 end as Income_Group
67 from loan_approved_clean;
68
69 -- 16. Find average loan amount for each income group.
70 select
71 case
72     when ApplicantIncome > 3000 then 'Low'
73     when ApplicantIncome between 3000 and 7000 then 'Medium'
74     else 'High'
75 end as Income_Group,
76 round(avg(LoanAmount),2) as avg_loan_amount
77 from loan_approved_clean group by Income_Group;
78
79 -- 17. Find applicants whose loan amount is greater than the overall average loan
amount.
80 select * from loan_approved_clean where LoanAmount > (select avg(LoanAmount) from
loan_approved_clean);
81
82 -- 18. Identify the property area with the highest average total income.
83 select Property_Area,
84 round(avg(ApplicantIncome + CoapplicantIncome),2) as avg_total_income from
loan_approved_clean
85 group by Property_Area
86 order by avg_total_income;
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87
88 -- 19. List all applicants whose income is above the average income of their education
category. Not Got
89 select * from loan_approved_clean where ApplicantIncome > ( select
avg(ApplicantIncome) from loan_approved_clean group by Education);
90
91 -- 20. Rank applicants based on total income (highest income rank = 1).
92 select * ,(ApplicantIncome + CoapplicantIncome) as Total_Income,
93 rank() over(order by (ApplicantIncome + CoapplicantIncome) desc
94 ) as Total_Rank
95 from loan_approved_clean;
96
97 -- 21. Show average loan amount per property area using a window function.
98 select * ,avg(LoanAmount) over(partition by Property_Area) as Avg_Loan_Amount from
loan_approved_clean;
99
100 -- 22. Calculate approval rate by education using grouping or window function.
101 select Education, count(*) as Total_Applicants,
102 sum(
103     case
104         when Loan_Status = 'Y' then 1 else 0 end
105     ) as Approved_Loan,
106 round(sum(case
107         when Loan_Status = 'Y' then 1 else 0 end
108     ) * 100 / count(*),2) as Approval_Rate
109 from loan_approved_clean
110 group by Education;
111
112 -- 23 Compare approval rate by credit history and education level to find which
combination performs best.
113 select Credit_History, Education, count(*) as Total_Applicants,
114 round(sum(
115     case
116         when Loan_Status = 'Y' then 1 else 0 end
117     ) * 100 / count(*),2) as Approval_Rate
118 from loan_approved_clean group by Education,Credit_History order by Approval_Rate
desc;
119
120 -- 24 Find the combination of property area and education with the highest approval
rate.
121 select Property_Area,Education,
122 round(sum(
123     case
124         when Loan_Status = 'Y' then 1 else 0 end
125     ) * 100 / count(*),2) as Approval_rate
126 from loan_approved_clean
127 group by Property_Area,Education
128 order by Approval_rate desc;
129
130 -- 25. Compare approval rate for self-employed vs non-self-employed applicants by
credit history.
131 select Self_Employed,Credit_History,count(*) as Total_Applicants,
132 round(sum(
133     case
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134         when Loan_Status = 'Y' then 1 else 0 end
135     )*100/count(*),2) as Approval_Rate
136 from loan_approved_clean group by Credit_History,Self_Employed
137 order by Credit_History,Approval_Rate desc;
138
139
```